

AptLens

Smart Recruitment Matching Platform

HOW TO RUN THE PROJECT

Fresh Lab PC - Clone, Configure, Database, Backend, Frontend, Login

Purpose

This guide is designed for a clean Windows lab PC. Follow the commands in order. It avoids editing committed source code and uses local runtime environment variables for the JWT key and demo administrator.

Repository	github.com/mhdmusaddiq/DevSphere-Smart-Recruitment-Matching-Platform
Branch	main
Verified main commit	3a77c235e32af240c38a4b56a89ec63ea089a8e4
Frontend / Backend	Angular 20.3.x / ASP.NET Core .NET 8

1. Before You Start - Lab PC Requirements

Install or confirm these items before cloning the repository.

Requirement	Recommended version	Check command	Why needed
Git	Current Git for Windows	git --version	Clone the public repository
.NET SDK	8.0.423 (project-pinned .NET 8 SDK)	dotnet --version	Build and run ASP.NET Core
Node.js	20.19+ or 22.12+	node --version	Run Angular 20
npm	Installed with Node.js	npm --version	Install frontend packages
SQL Server	SQL Server Express (SQLEXPRESS) or LocalDB	Get-Service *SQL* / sqllocaldb info	Local application database

Important database note

The repository default connection is Server=.

\SQLEXPRESS;Database=DevSphereDb;Trusted_Connection=True;TrustServerCertificate=True. The included helper script automatically falls back to (localdb)\MSSQLLocalDB when SQLEXPRESS is not available.

Quick prerequisite check

```
git --version
dotnet --version
node --version
npm --version
Get-Service -Name 'MSSQL$SQLEXPRESS' -ErrorAction SilentlyContinue
sqllocaldb info MSSQLLocalDB
```

2. Clone the Project from GitHub

Open Windows PowerShell and run:

```
cd C:\
mkdir AptLens-Lab -ErrorAction SilentlyContinue
cd C:\AptLens-Lab

git clone https://github.com/mhdmusaddiq/DevSphere-Smart-Recruitment-Matching-Platform.git

cd DevSphere-Smart-Recruitment-Matching-Platform
git checkout main
git pull origin main
```

```
git status
git log -1 --oneline
```

Expected result

git status should show a clean working tree on branch main. At the time this guide was prepared, main contained merge commit 3a77c235e32af240c38a4b56a89ec63ea089a8e4.

3. Trust the Local HTTPS Certificate

The Angular development proxy targets the backend at <https://localhost:7097>, so trust the .NET development certificate once:

```
dotnet dev-certs https --trust
```

Accept the Windows trust confirmation if prompted.

4. Configure Safe Local Runtime Values

In the PowerShell window that will run the backend, set these local-only environment variables:

```
$env:ASPNETCORE_ENVIRONMENT = "Development"

$env:JwtSettings__SecretKey = "AptLens-Lab-Development-Secret-Key-2026-DoNotUseInProduction"

$env:BootstrapAdmin__Enabled = "true"
$env:BootstrapAdmin__Email = "admin@aptlens.local"
$env:BootstrapAdmin__Password = "AptLens.Admin#2026!"
$env:BootstrapAdmin__DisplayName = "AptLens Administrator"
```

Do not commit these values

These are local demonstration credentials and a local JWT key. They are intentionally supplied at runtime rather than added to appsettings.json.

5. Database Setup

Option A - SQL Server Express (recommended because it matches the repository)

If SQL Server Express is installed as .\SQLEXPRESS, no connection-string change is needed.

```
Get-Service -Name 'MSSQL$SQLEXPRESS'
Start-Service -Name 'MSSQL$SQLEXPRESS' -ErrorAction SilentlyContinue
```

Option B - SQL Server LocalDB fallback

If SQLEXPRESS is not installed but LocalDB exists, run this in the same backend PowerShell window:

```
sqllocaldb start MSSQLLocalDB
```

```
$env:ConnectionStrings__DefaultConnection = "Server=(localdb)  
\MSSQLLocalDB;Database=DevSphereDb;Trusted_Connection=True;TrustServerCertificate=True"
```

Install/update EF command-line tool

```
dotnet tool install --global dotnet-ef --version 8.0.29
```

If PowerShell says dotnet-ef is already installed, run:

```
dotnet tool update --global dotnet-ef --version 8.0.29
```

Restore and create/update the database

```
dotnet restore .\backend\DevSphere.sln
```

```
dotnet ef database update `  
--project .\backend\src\DevSphere.Infrastructure\DevSphere.Infrastructure.csproj `  
--startup-project .\backend\src\DevSphere.Api\DevSphere.Api.csproj
```

Expected result

Entity Framework applies the repository migrations to DevSphereDb. On first backend start, the Admin, Employer and Candidate roles are seeded, and the configured bootstrap administrator is created.

6. Start the Backend API

Keep the same PowerShell window, because the local environment variables above exist only in that process.

```
dotnet run `
  --project .\backend\src\DevSphere.Api\DevSphere.Api.csproj `
  --launch-profile https
```

Expected backend addresses:

- Swagger: <https://localhost:7097/swagger>
- HTTPS API: <https://localhost:7097>
- HTTP fallback: <http://localhost:5129>

Keep this terminal open

The backend must continue running while you use the frontend. In Development mode, verification codes are also logged here.

7. Start the Angular Frontend

Open a SECOND PowerShell window:

```
cd C:\AptLens-Lab\DevSphere-Smart-Recruitment-Matching-Platform\frontend

npm ci
npm start
```

Open:

<http://localhost:4200>

The Angular dev server uses proxy.conf.json and forwards /api requests to <https://localhost:7097>.

8. Required Login Credentials

Use the following local demonstration accounts for the lab run.

Role	Email	Password	Availability
Admin	admin@aptlens.local	AptLens.Admin#2026!	Created automatically on first backend start after migration
Job Seeker	candidate@aptlens.local	AptLens.Candidate#2026!	Create once using Register -> Job Seeker

Employer	employer@aptlens.local	AptLens.Employer#2026!	Create once using Register -> Employer
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Password rule

AptLens requires strong passwords of at least 15 characters with upper-case, lower-case, number and symbol. The demo passwords above satisfy that rule.

Create the Job Seeker and Employer accounts on a fresh database

1. Open <http://localhost:4200/register>.
2. Choose Job Seeker, enter Candidate Demo, candidate@aptlens.local and AptLens.Candidate#2026!.
3. After registration, the Verify Email page opens.
4. Use the six-digit Development code shown in the backend terminal. If needed, wait for the resend cooldown and click Resend verification; the Development helper will show the local code.
5. Verify the account and sign in.
6. Repeat for Employer using employer@aptlens.local and AptLens.Employer#2026!.

9. Final Run Verification

Before submitting or demonstrating, confirm all of these:

- ☐ Backend terminal shows 'Now listening on: https://localhost:7097'.
- ☐ Swagger opens without an application startup error.
- ☐ Frontend opens at <http://localhost:4200>.
- ☐ No 'service unavailable' banner remains after successful API calls.
- ☐ Admin login opens the Admin area.
- ☐ Job Seeker login opens the Job Seeker area.
- ☐ Employer login opens the Employer area.
- ☐ Public Jobs page loads.
- ☐ Refreshing a protected route keeps routing correctly after authentication.

10. Fast Troubleshooting - Most Common Lab PC Failures

Problem	Cause	Fix
dotnet cannot find SDK 8.0.423	Required .NET SDK is missing	Install .NET 8 SDK 8.0.423, then reopen PowerShell.
Cannot connect to .\SQLEXPRESS	SQL Server Express instance is unavailable	Start MSSQL\$SQLEXPRESS or use the LocalDB fallback environment variable in Section 5.
JwtSettings:SecretKey must be supplied	Backend PowerShell did not receive the local JWT variable	Set JwtSettings__SecretKey again in the SAME terminal, then rerun the backend.
dotnet ef is not recognized	EF CLI tool not installed / PATH not refreshed	Install dotnet-ef 8.0.29; reopen PowerShell if necessary.
HTTPS certificate warning / proxy error	Local dev certificate not trusted	Run dotnet dev-certs https --trust, then restart backend/frontend.
npm ci fails	Wrong Node version or network/package cache issue	Confirm supported Node.js, then delete only frontend\node_modules if present and rerun npm ci.
Frontend says cannot reach AptLens	Backend is stopped or not on 7097	Keep backend terminal running with the https launch profile.
Login returns unauthorized after registration	Email has not been verified	Use the six-digit Development verification code first.
Email already exists when creating demo account	Demo user already exists in the local DB	Use the listed credentials to sign in instead of registering again.

11. Fastest Lab Demonstration Sequence

1. Start SQL Server/LocalDB.
2. Open backend PowerShell, set runtime variables, run migration if first time, start backend.
3. Open frontend PowerShell, run npm ci (first time only), then npm start.
4. Login as Admin with the bootstrap credentials.
5. Register/verify Job Seeker and Employer once if the fresh database has no demo accounts.
6. Demonstrate public jobs, role dashboards and core workflow.

12. Repository Facts Used by This Guide

- Database default: .\SQLEXPRESS / DevSphereDb / Windows trusted connection.
- Backend targets .NET 8 and the repository pins SDK 8.0.423.
- Backend HTTPS launch profile: https://localhost:7097; HTTP: http://localhost:5129.

- Frontend Angular packages are 20.3.31 with Angular CLI/build tooling 20.3.37.
- Frontend /api base URL is proxied to https://localhost:7097 during development.
- SMTP is disabled in committed local settings; Development mode provides local verification codes.
- Bootstrap Admin is runtime configuration only and is not hard-coded into the repository.

Submission note

This run guide is intentionally self-contained so a reviewer can clone the public main branch onto a fresh Windows machine and reproduce a local AptLens run without needing private project files.